

Global Dental Workforce and Basic Oral-Health Service Availability

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October 6, 2026

Micro and Macro Economics

Individual Research Paper

A problem and why it matters

The World Health Organization (WHO) estimated that approximately 3.5 billion people were affected by oral diseases, with three out of four affected people living in low- and middle-income countries. The burden falls disproportionately on disadvantaged populations (WHO, 2022c). In 2022, the World Health Assembly adopted a global strategy envisioning universal health coverage for oral health by 2030 (WHO, 2022a, 2022c).

Dental care depends on trained labor. This paper asks how dentist density, measured as dentists per 10,000 people, is associated with basic oral-health service availability across countries at different income levels. Workforce supply, training barriers, financing, and incentives to serve underserved populations are the economic questions behind this comparison.

My interpretation of the evidence

The analysis matches 171 countries using WHO dentist density, World Bank income groups, and three WHO 2021 service indicators: screening, urgent oral care, and basic restorative care (WHO, n.d.-b; World Bank, n.d.). The outcome measures general availability in public primary-care facilities, defined as reaching at least half of patients in need. Falling below that threshold does not mean no care exists (WHO, 2022b, p. 2).

Dentist supply and service availability had a modest relationship (Spearman rank correlation 0.306). Median density, or the middle value, was 2.97 where all three services were generally available, 0.76 where none met that criterion, and 0.72 where one or two did. It did not rise steadily across categories (Figure 1; Appendix Table A1).

Income was closely related to dentist supply. Median density rose from 0.06 in low-income countries to 0.31 in lower-middle-income, 2.33 in upper-middle-income, and 7.39 in high-income countries (Figure 2; Appendix Table A2). Within the two middle-income groups,

density had almost no relationship with availability. Income may therefore help explain the overall pattern, but this analysis cannot separate its influence from other factors.

I repeated the analysis using each country's latest dentist observation from 2021 or earlier. The correlation barely changed, from 0.306 to 0.305, retaining all 171 countries. I am less concerned that the original inclusion of 2022–2023 observations explains the relationship.

Some dentist observations remain older than 2021. Reporting differences, today's income groups, and the gap between availability and actual access remain concerns. Sources may count working or registered dentists and vary in completeness (WHO, n.d.-a). Country comparisons cannot prove cause and effect: income, health-system design, and spending could influence both supply and availability. National averages can hide local shortages. The analysis cannot show whether financing, provider location, or sharing tasks with other trained workers explains the differences (Appendix Table A3).

I expected dentist scarcity to be the main explanation. The findings challenged that expectation without establishing a better explanation.

Of the 99 countries reporting whether essential dental treatment was covered, those including it in their largest public scheme more often reported all three basic services as generally available (Table 1). The other 72 countries in the main sample lacked this financing response.

Table 1. Essential-curative benefit inclusion and basic service availability (2021)

Benefit included in largest public scheme	Countries	All three services generally available	95% Wilson interval
Yes	78	49 (62.8%)	51.7–72.7%
No	21	9 (42.9%)	24.5–63.5%

Source: calculations from WHO 2021 benefit-package and public-primary-care service indicators (WHO, n.d.-b, n.d.-c). Intervals describe individual proportions, not their difference.

The two-sided Fisher exact test gives $p = 0.135$, providing no strong statistical evidence of a difference. Patterns vary by income group, some subgroups are small, and median dentist density using observations through 2021 is higher where the benefit is included (2.23 versus 0.72). The test and intervals cannot separate financing from income or supply, or account for which countries reported data.

Policy options and my provisional judgment

I am considering three options:

1. **Expand dentist training and retention.** This could increase capacity, but requires money and time; graduates may still choose affluent or urban areas to work in.
2. Improve provider location and links with primary care. This could help existing providers reach underserved populations, but requires ongoing recruitment, retention, referrals, and public-sector support.
3. Expand public financing for essential dental care. This could make care more affordable, but additional demand may exceed provider capacity. The financing comparison highlights affordability and capacity together, without proving a provider shortage.

Recommendation. I recommend matching policy to workforce capacity and patients' barriers. The income differences, modest workforce relationship, and mixed financing results inform this tentative judgment; they do not show how many dentists are enough or which policy works best. Where local evidence confirms severe shortages, I would prioritize training, retention, and helping providers reach underserved areas. Where providers can deliver care but

cost is a major barrier, I would prioritize financing that reduces patients' costs, pays providers enough to participate, and encourages care in underserved areas.

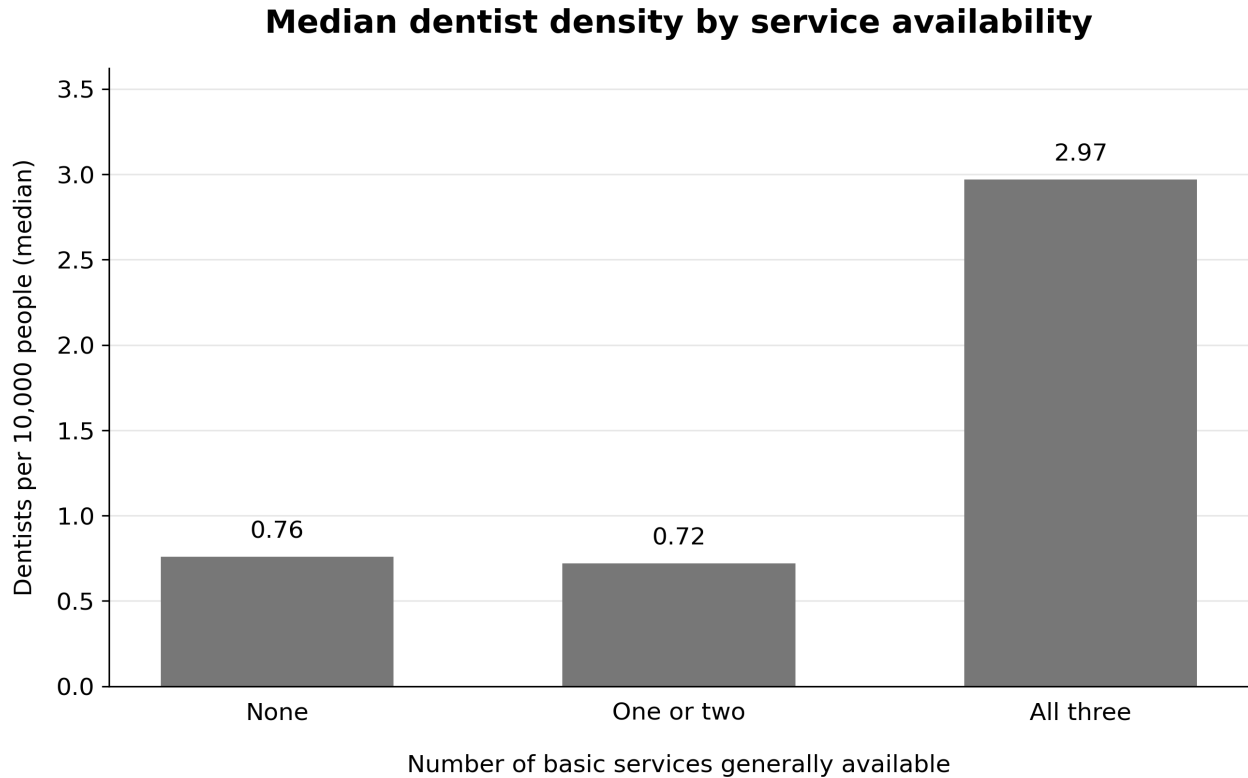
The strongest objection. My country comparisons cannot prove that financing improves access. Income or workforce capacity could explain both coverage and availability; the data cannot show which came first. Romania covers essential dental treatment in its largest public scheme, yet none of the three basic services is reported as generally available in public primary-care facilities (WHO, 2022b). Only 5% of dental spending was publicly funded in 2021 (OECD/European Observatory on Health Systems and Policies, 2023, p. 15). Listing a benefit therefore differs from reducing patients' costs. This share does not tell us who receives care or measure a reform's effect, and the availability measure may miss private treatment. Subsidies could raise prices or waiting times if providers cannot deliver more care; this is a theoretical risk, not my finding.

My response. My findings show a relationship, but do not prove cause and effect. My recommendation depends on local conditions. Public payments and patients' share of costs should make care affordable and encourage provider participation. Low payments can discourage participation, but my evidence does not establish this as Romania's cause. Incentives to serve underserved areas may also be needed. I would test reforms on a small scale, comparing costs with care received, affordability, unmet need, and oral health. I would reconsider if credible evaluations found no improvement after benefits expanded, or showed that workforce expansion or better provider distribution worked better at similar supply levels.

Bibliography

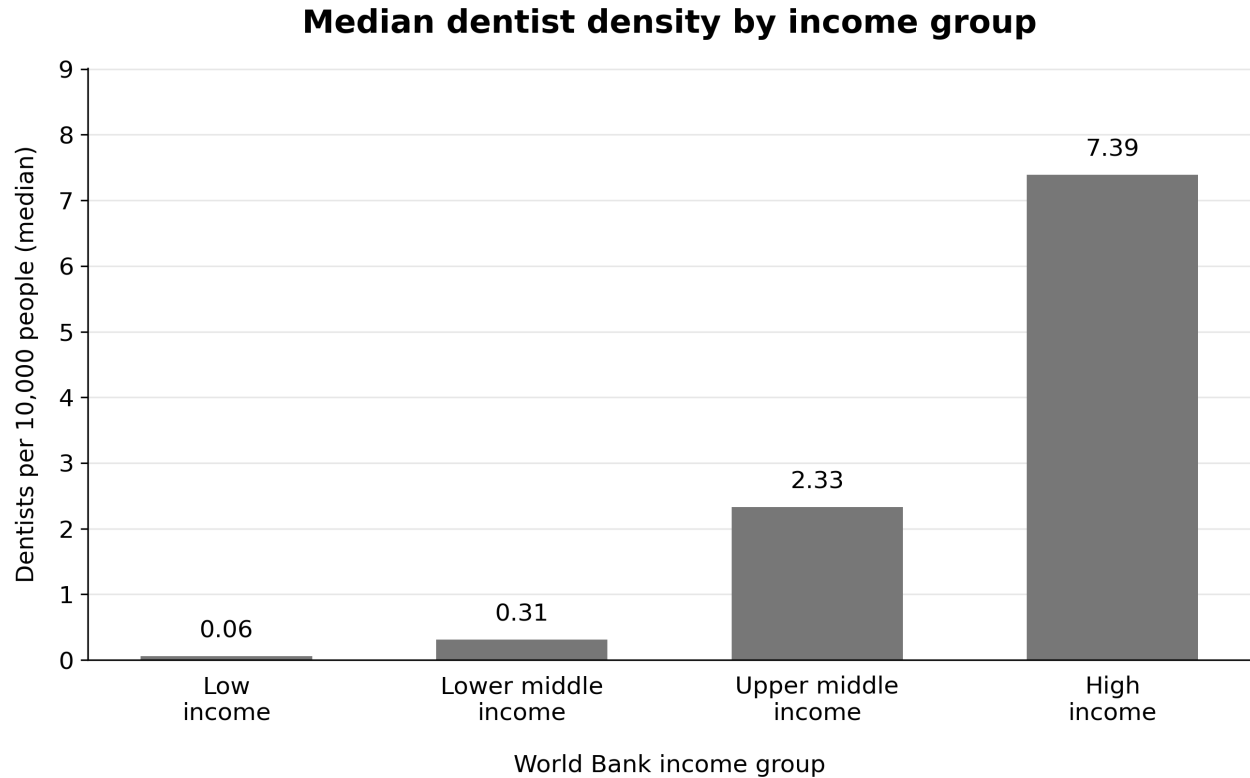
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Figure 1. Dentist density by service availability



Categories count services generally available in public primary-care facilities. Medians use workforce observations through 2023 and WHO 2021 service components. Source: author's calculations from WHO GHO/NCD CCS and World Bank data. Median comparisons do not establish causation.

Figure 2. Dentist density by income



Income groups are current World Bank API classifications, rather than historical 2021 groups.

Medians use workforce observations through 2023. Source: author's calculations from WHO workforce data and World Bank classifications.

Appendix

Appendix Table A1: Service availability and dentist density

Reported basic service availability	Median dentists per 10,000 people
None of the three services generally available	0.76
One or two services generally available	0.72
All three services generally available	2.97

Original workforce observations through 2023; outcome derived from WHO 2021 components. “Generally available” uses WHO's public-primary-care reach criterion.

Appendix Table A2: Income group and dentist density

World Bank income group	Median dentists per 10,000 people
Low income	0.06
Lower middle income	0.31
Upper middle income	2.33
High income	7.39

Source for Tables A1–A2: author's calculations from matched WHO workforce/service indicators and World Bank income classifications. These are original-baseline medians; the workforce-year sensitivity check is described in the main text.

Appendix Table A3: Limits of the analysis

What we cannot conclude	Why
More dentists cause better service availability	The analysis shows associations, not cause and effect.

What we cannot conclude	Why
Dentist supply does not matter	A weak national relationship can conceal local shortages or differences in how dentists work.
Financing, deployment, or task sharing explains the differences	These possible explanations have not yet been tested.
Patients actually receive affordable, accessible care	The outcome measures reported service availability in public primary care, not utilization, cost, travel distance, or quality.
A particular policy would improve access	The analysis has not evaluated the effects, costs, or tradeoffs of an intervention.

AI assistance disclosure

ChatGPT assisted with explanations of the analysis and policy options, table formatting, and revisions to wording and structure. Some passages originated in AI explanations shared during the drafting process. The policy preference remains provisional, and further source verification and methodological checks are pending.